

Abhishek Thakur

Robotics Engineer



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📍 Schmalkalden, Germany
(Open to Relocation)

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🎓 EDUCATION

Masters in Mechatronics and Robotics

Hochschule Schmalkalden

10/2023 – Present

Schmalkalden, Germany

Bachelors in Mechanical Engineering

K. K. Wagh Institute of Engineering

Education and Research

06/2017 – 05/2021 | Nashik, India

📖 COURSES

SUAS Premier

Hochschule Schmalkalden

09/2024 – 12/2024

Schmalkalden, Germany

Professional STEM (MINT) skill development and intensive language program focus on German labour market.

🌐 LANGUAGE

English - Bilingual ● ● ● ● ●

German - Intermediate ● ● ● ● ●

Hindi - Native ● ● ● ● ●

🧠 TECHNICAL SKILLS

- **Programming:** C++, Python, Rust, MATLAB
- **Robotics & Simulation:** ROS2, ROS, Gazebo, RViz, MoveIt2, microROS, SLAM, Navigation2, Sensor Fusion
- **Control & Embedded:** STM32, Arduino, Raspberry Pi, PLCs, CAN bus, Ethernet, TCP/UDP, Serial Communication, RTOS, PID Control
- **AI & Perception:** OpenCV, TensorFlow, PyTorch, Keras, Computer Vision, Machine Learning, Deep Learning
- **Tools & DevOps:** Linux, Docker Containerization, Git, GitHub Actions/GitLab, Cmake, Continuous Integration
- **Mechanical & Prototyping:** SolidWorks, AutoCAD, PTC Creo, PCB Design, 3D Printing, CNC Machining

📁 EXPERIENCES

Eviotec Technologieentwicklung GbR

Project Assistant (Projektmitarbeiter)

04/2025 – 08/2025 | Schmalkalden, Germany

- Led end-to-end development of an industrial-grade automated system, from concept research to prototype, integrating embedded control and custom hardware.
- Evaluated and tested hardware components using mathematical modeling, computer simulation, and real-world validation.
- Developed a functional control prototype on STM32 microcontroller, integrating sensors, actuators, and custom electronics.
- Designed mechanical components and system layout using CAD tools, ensuring manufacturability and safety compliance.
- Collaborated via Taiga portal and LaTeX-based documentation, analogous to Confluence workflows.
- Prepared technical documentation and delivered presentations in German to stakeholders.

Prime Graphite Industry

Production Planning Engineer

12/2022 – 07/2023 | Nashik, India

- Optimized CNC manufacturing workflows with automation scripts, reducing CAD processing time by 62%.
- Enhanced fixture design for precision manufacturing.
- Coordinated cross-department processes from design to production, applying principles of industrial automation.

CERTIFICATIONS

Robotics

NPTEL- IIT Kharagpur

Python Data Structures

University of Michigan

Aerial Robotics

University of Pennsylvania

Machine Learning

Stanford Online

Design Thinking for Innovation

VOLUNTEERING

Rotaract Club of Nashik Nine Hills

Member

Nashik, India

Organized community service activities, charity events, and social service initiatives

DEGIS

Volunteer

EXTRA-CURRICULAR

Active member of hiking and conversation clubs in Germany.

PROJECTS

LLM/NLP Controlled UR5e Robot System

(ROS2, Python, ROSA, LLMs, AI agents)

Ongoing

- Designed a voice- and text-command interface for the UR5e (Universal Robot) robotic arm using locally hosted Large Language Models integrated with ROS2.
- Implemented motion planning and execution via MoveIt2 with natural language task parsing.
- Containerized the project in Docker for reproducible deployment.
- Enabled autonomous execution of multi-step manipulation tasks, improving human-robot interaction efficiency by ~40% in lab testing.

MicroROS Digital Twin Robotics Framework

(ROS2, microROS, C++, Python, MoveIt2, RViz, Gazebo)

- Designed and implemented a custom inverse kinematics solver using Jacobian-based central difference computation for real-time robotic arm control without MoveIt.
- Integrated the system into Edrys Lab (University of Freiburg) research platform, enabling remote robotics experiments for educational use.
- Developed a high-fidelity digital twin of the ASRS robot in Gazebo with enhanced URDF, physics tuning, and motion planning via MoveIt2.
- Planned microROS-based synchronization between digital twin and physical robot, enabling distributed real-vs-simulation experiments.
- Research paper in preparation

Advanced SLAM on Unitree Go1 Pro

(ROS2, C++, Nav2, SLAM Toolbox, Sensor Fusion)

Ongoing

- Integrated multi-sensor fusion (Lidar + IMU + stereo vision) for robust indoor navigation.
- Developed a real-time mapping pipeline with loop closure and localization recovery.
- Achieved <5 cm localization error in dynamic environments during test runs.

Disaster Response Robot (Bachelor Thesis)

(OpenCV, Python, UGV Platform)

- Built an unmanned ground vehicle for search-and-rescue operations in hazardous environments.
- Implemented real-time human detection using computer vision, achieving >90% detection accuracy in varied lighting conditions.
- Integrated wireless control and live video streaming for remote operation.

Academic Seminar: Anthropomorphic Prosthetic Limbs with Myoelectric Control System

- Presented research on robotic hands with haptic feedback, focusing on myoelectric control systems for anthropomorphic prosthetics.